Total No. of Questions :6]

SEAT No. :

P63

[Total No. of Pages :2

OCT. -16/BE/Insem. - 117
B.E. (Mechanical)
ENERGY AUDIT & MANAGEMENT
(2012 Course) (Semester - I) (Elective-I) (402044 A)

Time : 1 Hour] *[Max. Marks :30*

Instructions to the candidates:

1) ***Neat diagrams must be drawn wherever necessary.***
2) ***Figures to the right indicate full marks.***
3) ***All questions carry equal marks.***
4) ***Use of logarithmic tables slide rule, Mollier charts, electronic pocket calculator and steam tables is allowed.***
5) ***Assume suitable data if necessary.***

Q1) a) Explain Need of Energy Management in India. **[5]**

b) Explain the following term

i) Primary and Secondary Energy. **[3]**

ii) Commercial and Non Commercial Energy. **[2]**

OR

Q2) a) Explain with neat sketch Global Warming Potential. **[5]**

b) Explain Need of Renewable Energy and Non Renewable Energy with respect to their energy efficiency. **[5]**

Q3) a) List out Different Instruments used for ice plant Energy Audit and Explain any Two. **[5]**

b) Describe Energy Conservation Opportunity in Cooling Tower. **[5]**

OR

Q4) a) Explain Full (Detailed) Energy Audit. **[5]**

b) Describe Energy Conservation Opportunity in Furnace. **[5]**

P.T.O.

Q5) a) Explain Net Present value Financial Analysis Method. **[5]**

b) A Proposed Project requires an capital (initial) Investment of Rs.20,000.**[5]**

Investment Saving in Year	Rs.20,000. Cash Flow
1	6,000
2	5,500
3	5,000
4	4,500
5	4,000
6	4,000

The Cost of capital (discount rate) K, for the firm is 8%. Calculate IRR.

OR

Q6) a) Investment for an Energy Proposal is Rs.10.00 lakhs. Annual savings for the first three years is 150,000, 200,000 and 300,000. Considering cost of capital as 10%, what is the Net Present Value of the Proposal? Will Project meet firm Expectations. **[5]**

b) Calculate the internal rate of return for an economizer that will cost Rs.500,000, will last 10 years and will result in fuel savings of Rs.150,000 each year. Find the i that will equate the following:

Rs.500,00 = 150,000 x PV (A = 10 years, i=?) **[5]**

E E E